

Advanced (Habanero)

Q1. Factor the expression: $x^2 + 5x + 4x + 20$

- (A) $x(x + 5) + 4(x + 5)$
- (B) $x(x + 4) + 5(x + 4)$
- (C) $x(x + 4) + 5(x + 5)$
- (D) $x^2 + 9x + 20$
- (E) $x(x + 5) + 4(x - 5)$

Q2. Factor: $x^2 + 7x + 2x + 14$

- (A) $x(x + 7) + 2(x + 7)$
- (B) $x(x + 2) + 7(x + 2)$
- (C) $x(x + 2) + 7(x - 2)$
- (D) $x^2 + 9x + 14$
- (E) $x(x + 7) - 2(x - 7)$

Q3. Factor the expression: $3x^2 + 12x + 4x + 16$

- (A) $3x(x + 4) + 4(x + 4)$
- (B) $3x(x + 4) - 4(x - 4)$
- (C) $3x(x + 2) + 4(x + 4)$
- (D) $3x^2 + 16x + 16$
- (E) $3x(x + 4) - 4(x + 4)$

Q4. Factor by grouping: $2x^2 + 10x + 3x + 15$

- (A) $2x(x + 5) + 3(x + 5)$
- (B) $2x(x + 3) + 5(x + 3)$
- (C) $2x(x + 3) - 5(x - 3)$
- (D) $2x^2 + 13x + 15$
- (E) $2x(x + 3) + 5(x - 3)$

Q5. Verify the factoring: $x^2 + 6x + 8 = ()$

- (A) $x(x + 2) + 4(x + 2)$
- (B) $x(x + 4) + 2(x + 4)$
- (C) $x(x + 6) + 2(x + 2)$
- (D) $x(x + 8) + 6(x - 2)$
- (E) $x(x + 8) - 2(x - 4)$

Q6. Factor the expression: $x^2 + 4x + x + 4$

- (A) $x(x + 4) + 1(x + 4)$
- (B) $x(x + 1) + 4(x + 4)$
- (C) $x(x + 1) + 4(x - 1)$
- (D) $x^2 + 5x + 4$
- (E) $x(x + 4) - 1(x - 4)$

Q7. Factor the expression: $2x^2 + 6x + 4x + 12$

- (A) $2x(x + 3) + 4(x + 3)$
- (B) $2x(x + 4) + 3(x + 4)$
- (C) $2x(x + 2) + 6(x + 2)$
- (D) $2x^2 + 10x + 12$
- (E) $2x(x + 3) - 4(x - 3)$

Q8. Factor by grouping: $x^2 + 2x + 5x + 10$

- (A) $x(x + 2) + 5(x + 2)$
- (B) $x(x + 5) + 2(x + 5)$
- (C) $x(x + 5) - 2(x - 5)$
- (D) $x^2 + 7x + 10$
- (E) $x(x + 5) + 2(x - 5)$

Open Ended Questions (FRQ)

Q9. Factor the expression $3x^2 + 12x + 4x + 16$ and verify your answer by multiplying the factors.

Q10. Factor by grouping: $2x^2 + 8x + 5x + 20$ and simplify the result.

Chef's Tips

- Read each question carefully before answering.
- Check your work and verify your answers.
- Show all work and explain your reasoning.